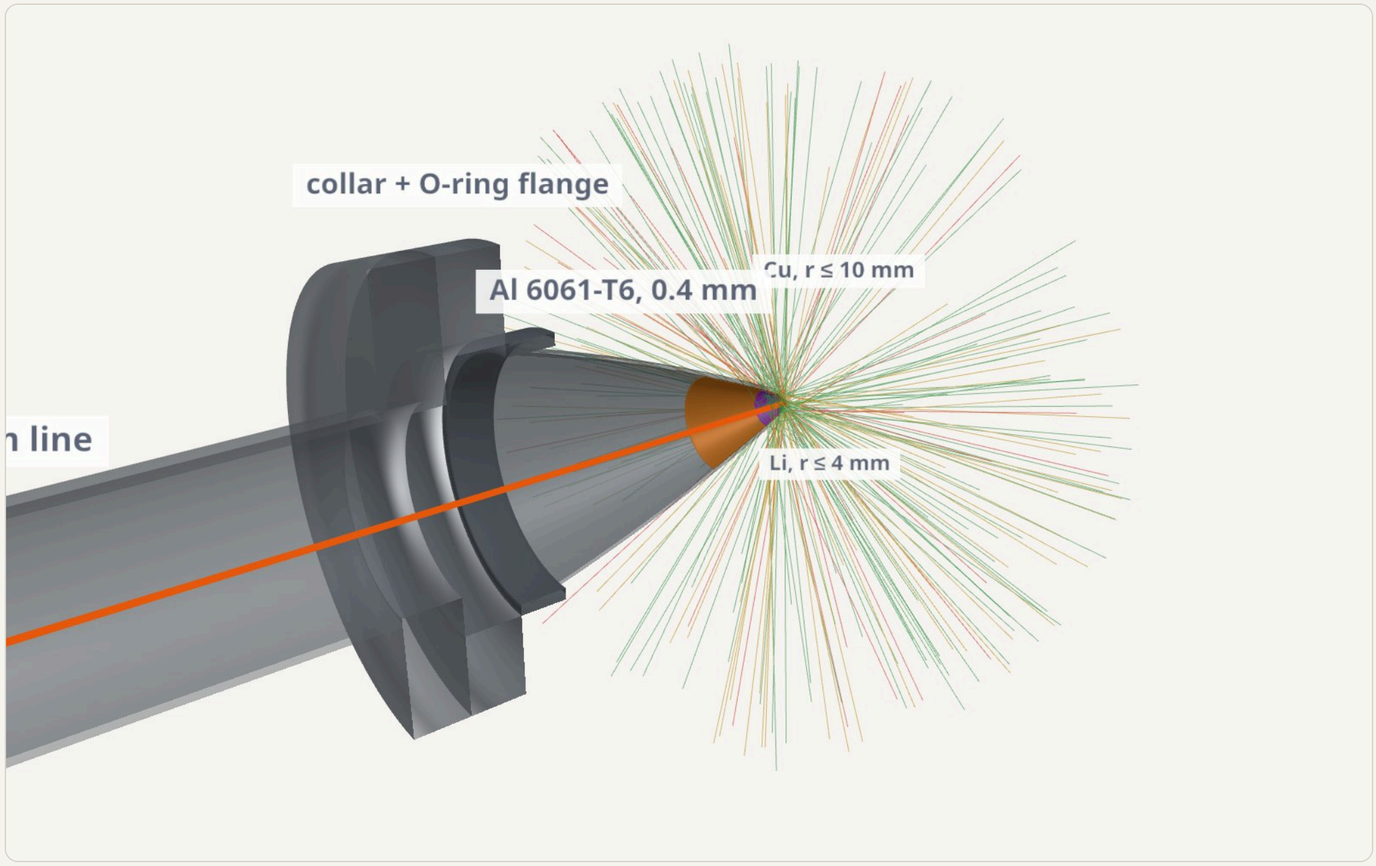
Your ogive, in Al, with Cu only where the beam stops

Same shape as your 27 Oct CAD. Al 6061-T6 0.35–0.4 mm · ≥ 12 µm Cu over r ≤ 10 mm · 20 µm Li over r ≤ 4 mm · both films laid from the beam axis



11.5° vs 18.5°
σ68 of the opening angle, against the Cu ogive (Geant4 L12/L11)

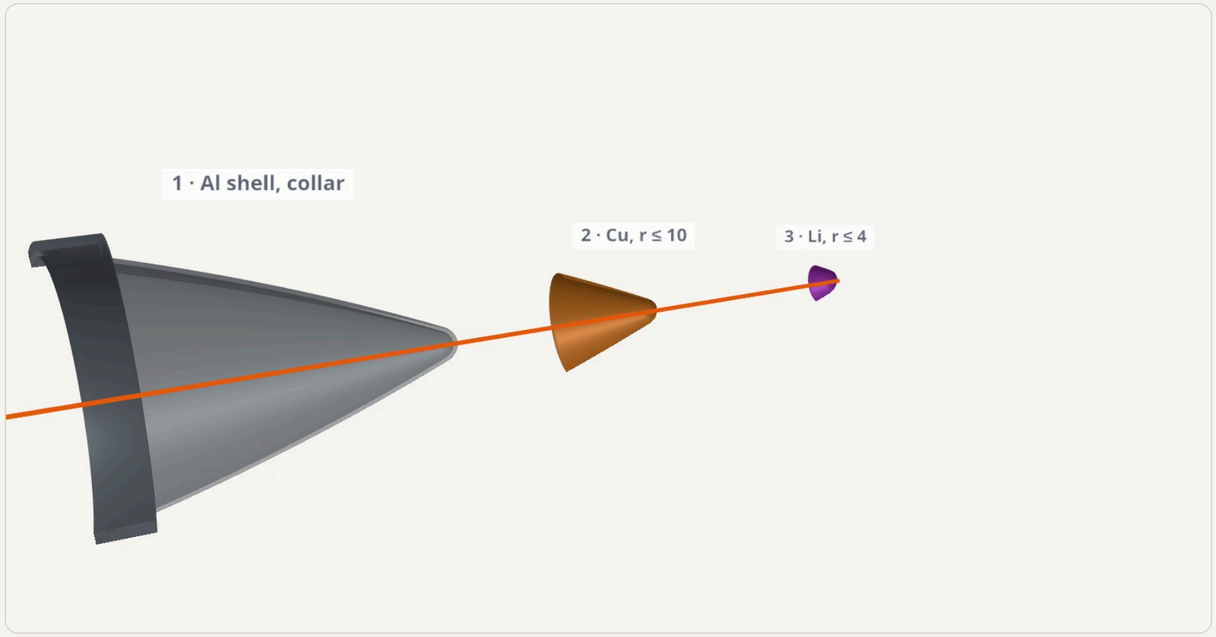
×2.5–3.5
faster to 3σ at the ATOMKI strength at the same current (Al 0.4–0.3 mm)

4 % vs 22 %
of the background from γ converting in the shell

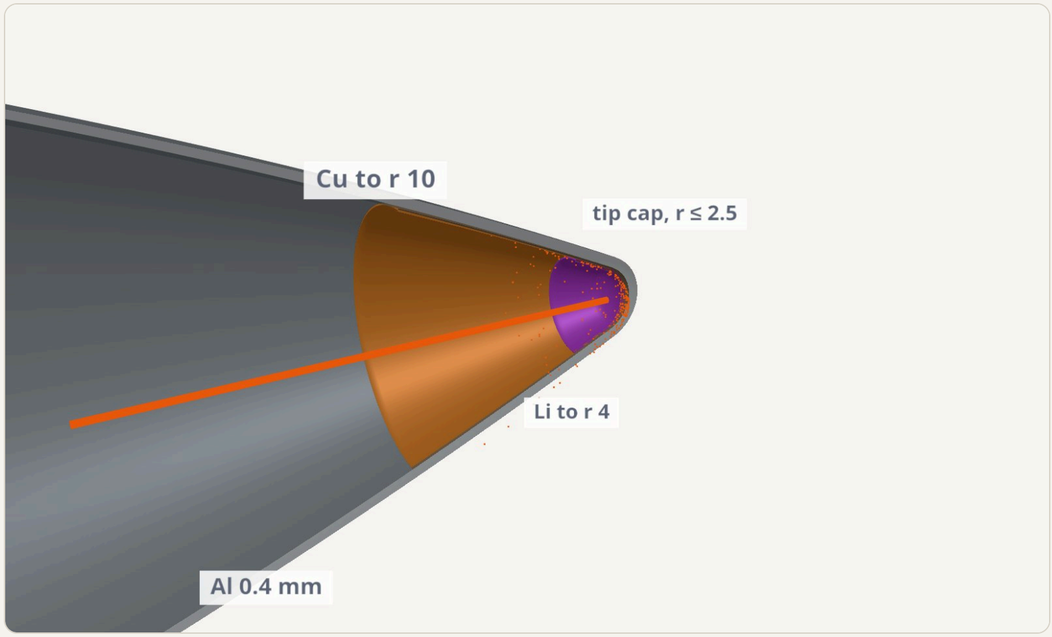
The leptons cross 0.4 mm of Al instead of 0.2 mm of Cu. The Cu is only a beam stop: the protons end in it, never in the Al.

Three layers, the thin ones only at the tip

A σ 2 mm beam puts 99.9 % of its protons inside r 7.4 mm, so nothing needs to be coated beyond r ≈ 10 mm



Pulled apart along the beam: the Al shell, the Cu cap, the Li cap (thin layers drawn ×20–30).



The tip, cut open: Li (purple) on Cu (orange) on Al; dots are where a σ 2 mm beam lands.

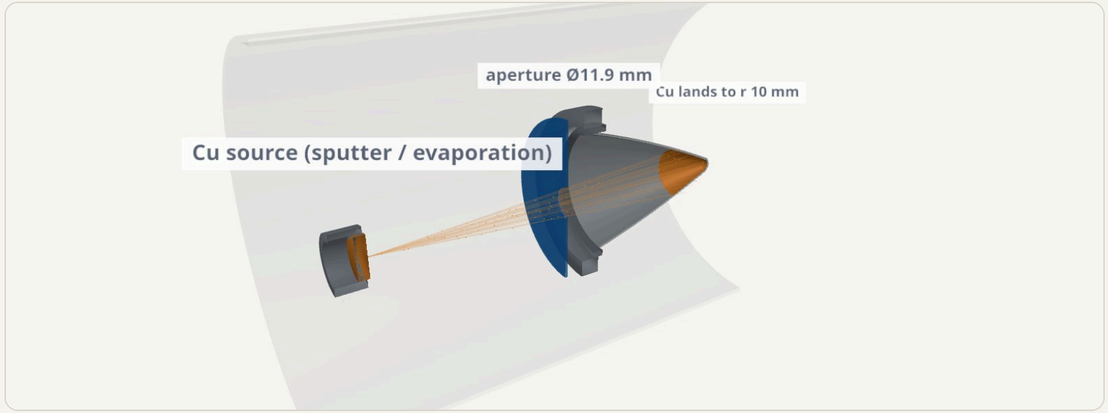
layer	how thick	where	why
Al 6061-T6	0.35–0.4 mm	the whole shell + a 2.5 × 10 mm collar	holds 1 bar (×10 margin at 0.4); stops the Cu K x-rays
Cu	≥ 12 µm along the beam	r ≤ 10 mm (the tip and first ~19 mm of flank)	stops the protons (10.7 µm needed); Li must not touch Al
Li	20 µm along the beam	r ≤ 4 mm, inside the Cu	the target; r > 4 sits up to 9 mm upstream of the tip

How it is built: one vacuum system, the source on the axis

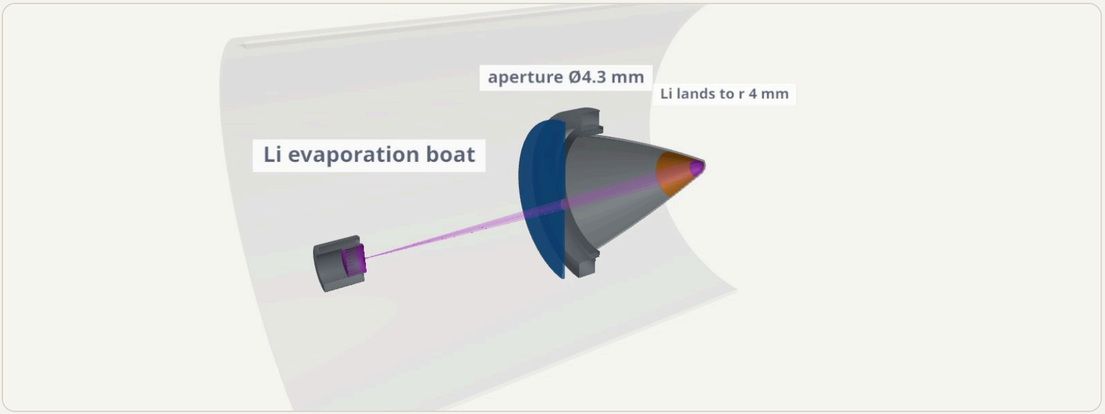
the open base faces the source 150–400 mm away; aperture plates decide where each film ends



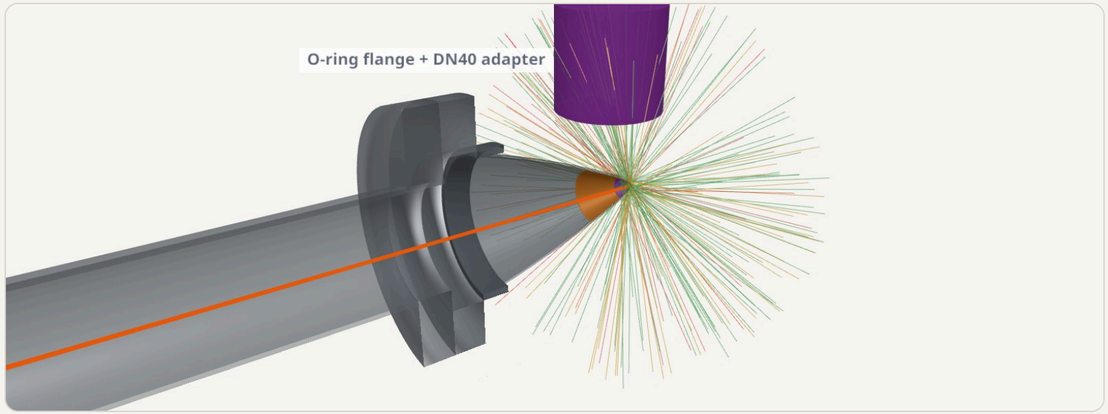
1 · Shell. Machine or spin the Al ogive, 0.35–0.4 mm, with a thick collar at the base for the joint (63 mm behind the tip, out of the arms' view).



2 · Cu. Ar-ion etch the inside, then sputter or evaporate 12–16 µm of Cu through a Ø12 mm aperture: it lands to r ≈ 10 mm. 1–2 h at 0.1–0.3 µm/min.



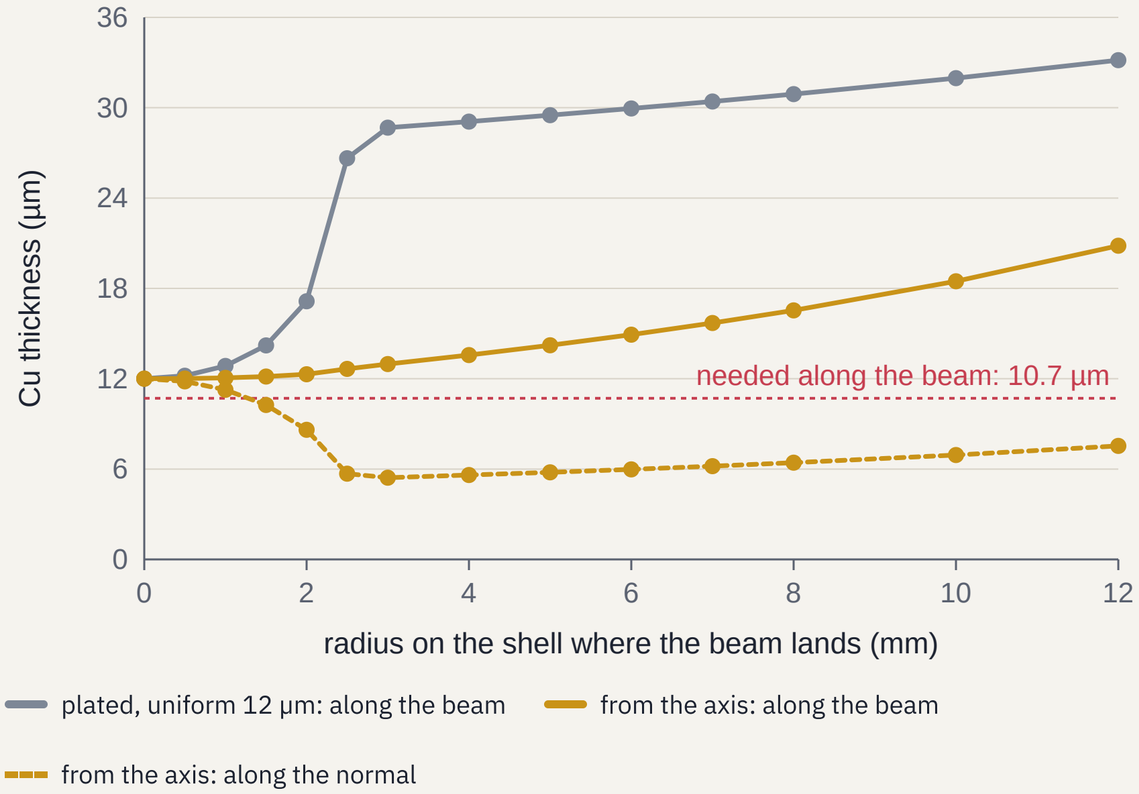
3 · Li. Without breaking vacuum (or under Ar), evaporate 20 µm of Li through a Ø4 mm aperture: it lands to r ≈ 4 mm, inside the Cu. A witness coupon records both thicknesses.



4 · On the line. Clamp the collar on an O-ring flange, adapter to DN40, He leak test. Minutes of PIGE at 1.10 MeV: no Al lines, 478 keV present.

Why lay both films from the axis

a line-of-sight coat lands ∝ cos(incidence), the same factor by which the beam's path through it grows



On the 65° flank a film of fixed normal thickness is ×2.4 longer along the beam. For the Cu that is only wasted metal; for the Li it is a problem: the protons leave it below 441 keV and the 17.64 MeV line grows.

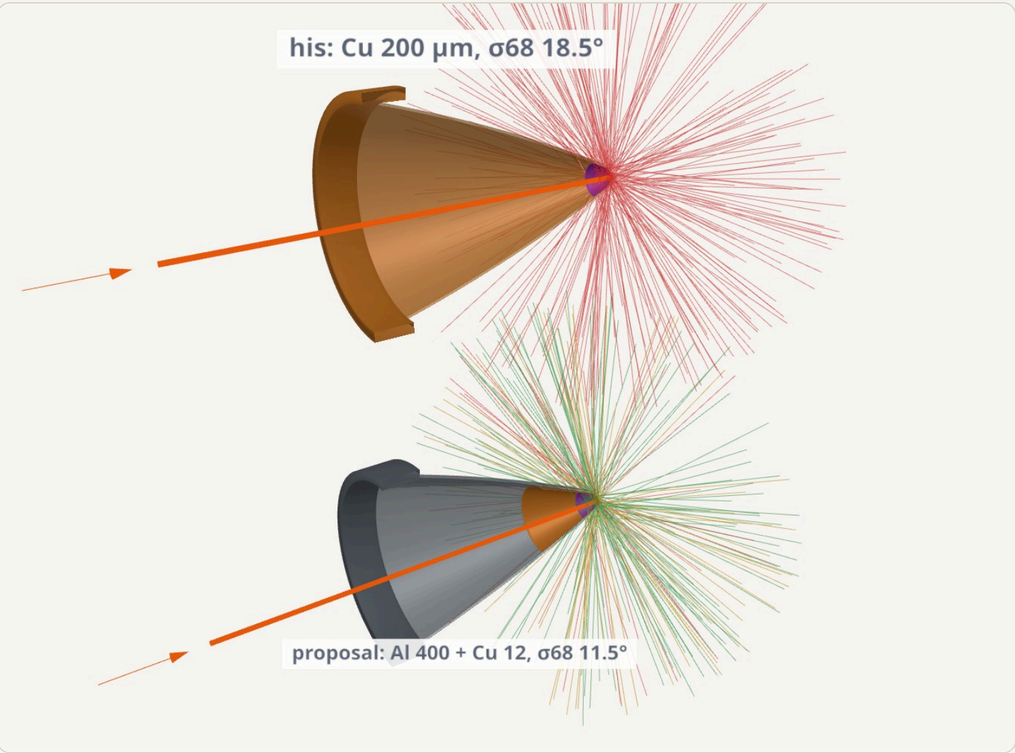
17.64 MeV share of the γ_0

Li film	1.10 MeV	1.03 MeV
same thickness along the normal	29.3%	70.7%
laid from the axis, r ≤ 4	0.9%	1.4%

From the axis both films come out constant along the beam: 12–14 µm of Cu to r 8 mm, ~20 µm of Li everywhere. 1.03 MeV then costs only ~15 % against 1.10.

Against the Cu ogive: same shape, the metal changes

Geant4 L11/L12, big plastics, days to 3σ at the ATOMKI strength per μA (statistics only)



Top: Cu 200 μm, every leg crosses the Cu (red). Bottom: Al 0.4 mm + Cu liner, legs by Al crossed.

	σ68	days/μA 1.10	1.03 MeV	EPC	MM x-rays /arm /μA
his Cu ogive, 200 μm	18.5°	1.13	1.29	22%	~0.1 %
Al 300 + Cu 12	10.5°	0.33	0.38	3%	12 %
Al 400 + Cu 12	11.5°	0.44	0.52	4%	2.5 %
Al 300 + Ti 15	9.8°	0.34	0.40	2%	0

The price is Cu K x-rays (8 keV) from where the beam stops. Al absorbs them hard: each 0.1 mm divides what reaches the Micromegas by ~3.7. Hence **0.4 mm**: 2.5 % occupancy per arm per μA, so up to ~4 μA.

Ti instead of Cu (15 μm) makes no x-ray that escapes and gives σ68 9.8°, but takes up hydrogen.

At full current your Cu ogive is not slow (no x-rays); the Al one wins on what matters for a claim: resolution, ×6 less EPC, ×2.8 fewer trigger legs.

YOUR VIEW?

What we would like to know

1 · The **beam spot** at the target, σ and halo: it sets the Cu and mask radii (we assume σ 2 mm).

2 · Can the **thin-film lab** ion-etch, then sputter or evaporate ~15 µm of Cu inside the ogive from the axis, and Li through a mask, ideally without breaking vacuum?

3 · Can the workshop make the shell in **Al 6061-T6 at 0.35–0.4 mm** with a collar (not 0.2 mm)?

4 · What **current** do you expect? At ≤ 1 µA the x-rays never matter.

5 · Would **Ti** (15 µm) be as easy as Cu?

Interactive 3D, and every number:

dylan-neff.web.cern.ch/facilities/lnl-target.html
(tab “Al ogive + Cu liner”)

The full study: notes/lnl-cu-liner